

Shay Mordechai

Security Researcher | Systems, Network Internals & Embedded Security

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Summary

Offensive Security Researcher specializing in Network Protocol Security, Cloud Architectures, and vulnerability discovery. Experienced in AI-augmented threat hunting, dynamic analysis, and developing bypass techniques against complex environments (V8, Compilers) to uncover zero-day logic flaws.

Professional Experience

Independent Security Researcher & Architect | *MyLeads AI* | Jan 2026 – Present

- **Zero-Trust Infrastructure:** Engineered a "Dark Server" AWS architecture (0 inbound ports), routing traffic via Cloudflare Tunnels and Rootless Podman.
- **Red Teaming & Pen-Testing:** Audited architectures, securing API endpoints against advanced network TTPs and bypass attempts.
- **Wasm DLP Firewall:** Engineered a Rust WebAssembly filter for Envoy Proxy to redact sensitive egress traffic, preventing Information Disclosure.

Application Security Intern | *Israel Tax Authority* | Aug 2024 – Jul 2025

- **Defense-in-Depth:** Managed Threat Modeling and access controls. Hardened IBM DataPower gateways via strict JSON schemas.
- **AppSec & Remediation:** Integrated Checkmarx SAST/SCA into 15+ CI/CD pipelines. Triaged findings and trained teams to mitigate vulnerabilities.

IDF Combat Soldier (Hesder) | *Combat Intelligence Collection Corps* | Aug 2016 – Apr 2018 - Executed covert reconnaissance and visual anomaly detection for actionable operational target profiling.

Education

B.Sc. Computer Science | *Jerusalem College of Technology* | 2021-2025 | GPA: 89

- **Coursework:** Network Analysis (97), InfoSec (90), Advanced Malware Analysis (IoCs, Yara - In Progress).
- **Reverse Engineering (93):** Conducted static/dynamic analysis of x86/x64 binaries and mobile components via **IDA//x32dbg** to reverse-engineer logic and bypass anti-debugging.

Technical Projects & Research

- **Hardware Guard (Rust/ESP32):** Developing a bare-metal gateway. Implemented DPI to enforce strict TTL policies and mitigate L7 attacks (DNS Rebinding).
- **SBCL Compiler Vuln Research:** Executed **pre-deployment red-teaming** and vulnerability research via AI-augmented semantic fuzzing (Python). Discovered scope bypasses to assist in **model hardening**, reporting 3 Zero-Days.
- **V8 Engine Internals:** Published analysis on V8 JIT vulnerabilities, utilizing **detours**, runtime hooking, and code manipulation to bypass CFI via RWX memory.
- **DNS Rebinding Research:** Discovered local interface bypasses. Authored rigorous exploit reports, validated by TP-Link & Palo Alto PSIRT.

Technical Skills

- **Security & Research:** Reverse Engineering, Exploit Development, Fuzzing, Runtime Code Manipulation, Bug Bounty.
- **Languages & Low-Level:** Python, C/C++, x86 Assembly, Bash, Rust, OS Internals.